

CONFLUENCE × OnCo: An Evidence-Gated Dynamical Framework for Integrating Oncology Knowledge Graphs with Adaptive Cancer-State Models

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Independent computational research / Project Confluence (GitHub cloudynirvana)

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Abstract

Cancer research now produces knowledge graphs, multi-omic assays and dynamical simulators in parallel [10,53,61]. The failure mode is collapsing those layers. This study asks whether a provenance-controlled pipeline can keep the layers apart while still allowing testable predictions. Scientific success is a staged chain — traceability, mathematical validity, identifiability, out-of-sample prediction, experimental falsification — not disease eradication [43-52]. PR #9 is architectural evidence for the gates, not therapeutic efficacy [11,56]. GLOBOCAN 2024 estimates, published 2026, report about 20.6 million diagnoses and 9.8 million deaths; that statistic is context, not a CONFLUENCE parameter [1,2,64].

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<https://github.com/cloudynirvana/project-confluence>

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Revision: 21 September 2026 — Problem Statement, Justification of the Study, and Significance of the Study added before Methods.

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Status: Architectural and methodological findings. Not a clinical result.

Citation style: numbered Vancouver [n] matching the References list and the public thesis page. See docs/CITATION_STYLE.md.

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This manuscript expands ONCO_CONFLUENCE_THESIS_FINDINGS.md, docs/ONCO_CONFLUENCE_ONTOLOGY_SPEC.md, and docs/ONCO_ADAPTER.md. It does not add wet-lab measurements, patient-level results, or invented identifiers.

Cancer research now produces knowledge graphs, multi-omic assays and dynamical simulators in parallel [10,53,61]. The failure mode is collapsing those layers. This study asks whether a provenance-controlled pipeline can keep the layers apart while still allowing testable predictions. Scientific success is a staged chain — traceability, mathematical validity, identifiability, out-of-sample prediction, experimental falsification — not disease eradication [43-52].

CONFLUENCE v2 is a frozen 15-dimensional computational cancer-state model with an optional connectome-style controller [11,58,60]. OnCo is a public, cited oncology knowledge graph [10,61]. They are complementary only if knowledge is not written into parameters [57,58]. Pull request #9 shipped a read-only OnCo adapter and nine refusal tests. It is architectural evidence for the gates, not therapeutic efficacy [11,56].

GLOBOCAN 2024 estimates, published 2026, place the contemporary burden at about 20.6 million diagnoses and 9.8 million deaths [1,2,64]. That statistic motivates computational oncology; it is not a CONFLUENCE parameter [1,59]. This document records the gates, the P0 adapter, the LDHA / p_{lactate} refusal, and the honesty rule that OnCo confidence is not P(H) [10,11,36,63]. It does not claim a cure, a dose, a clinical decision-support system, or an identified Θ [3,56].

Keywords

computational oncology; knowledge graphs; dynamical systems; evidence gates; identifiability; tumour heterogeneity; tumour microenvironment; OnCo; CONFLUENCE; research-only; not a medical device

Introduction

Oncology knowledge graphs name genes, diseases, ideas and citations [10,53,61]. Dynamical simulators ask how a state moves under control [8,39,43,44]. Mixing the two without provenance treats a web page as an identified parameter [45-52,57]. The problem stated on the public thesis page, and restated here, is: how can heterogeneous oncology knowledge and molecular observations be incorporated into a cancer dynamical model without collapsing evidence, mechanism, parameterisation and prediction into unsupported assumptions [57,59]?

GLOBOCAN 2024 estimates, published 8 July 2026, report about 20.6 million diagnoses and 9.8 million deaths across 34 cancer types in 186 countries [1,2,64]. Female breast cancer accounted for about 2.43 million new cases in that estimate series [1,2]. Those figures continue a modelled GLOBOCAN series previously reported for 2022 and 2020 [54,55]. WHO states that many cancers can be cured if detected early and treated effectively, while access remains uneven [3]. That sentence is a policy statement about staged, treatable disease — not a claim that CONFLUENCE or OnCo cures patients [3,10,11,56]. The WHO global status report on cancer 2026 is cited as health-system context only [4]. Nigeria GLOBOCAN 2022 estimates (127,763 new cases; 79,542 deaths; breast 32,278) are setting context only; they are not 2026 incidence [5].

Cancer is organised as a set of acquired capabilities — the hallmarks — rather than as a single targetable defect [13-15]. Tumours evolve clonally and adapt under therapy [7,16,17]. Intra-tumour heterogeneity, branched evolution and therapy resistance are documented empirical features of that process [18-21]. Multi-omics increase resolution without automatically producing a causal model [20,43,49]. The tumour microenvironment recruits stromal and immune cells that regulate progression, exclusion and metastasis [22-27]. Mathematical oncology supplies in-silico

laboratories that couple those layers as dynamical systems [8,39]. Adaptive therapy treats treatment as a process under selection rather than as a fixed maximum-tolerated pulse [9,40,41]. The gap this thesis addresses is connecting knowledge to dynamical hypotheses without dropping provenance [10,11,57].

TNBC is used as a test case because NCI describes it as roughly 15% of breast cancers, typically faster-growing and more recurrent, and heterogeneous [6]. Independent reviews describe the same clinical heterogeneity and recurrence pattern [28-31]. That description is not a CONFLUENCE parameter [6,56,63].

1. Problem Statement

How can heterogeneous oncology knowledge-graph records and molecular observations be incorporated into a frozen cancer dynamical model without collapsing knowledge, evidence, mechanism, parameterisation, and prediction into unsupported assumptions [10,45-53,57,61]?

That is the research problem. It is a computational and medical-methods problem. It is not a claim to treat, dose, or cure patients [3,56].

Cancer research now produces knowledge graphs, multi-omic assays, and dynamical simulators in parallel [10,20,43,44,53,61]. OnCo names genes, diseases, ideas, and citations [10,61]. CONFLUENCE v2 asks how a frozen 15-dimensional state moves under an in-silico controller [11,58,60]. Mixing the two without provenance treats a web page, an OnCo confidence score, or a legacy numeric map as an identified parameter Θ [45-52,57,63]. The conversion ladder — Knowledge $\rightarrow$ Evidence $\rightarrow$ Hypothesis $\rightarrow$ Mechanism $\rightarrow$ Parameter $\rightarrow$ Prediction $\rightarrow$ Experiment $\rightarrow$ new Evidence — has a failure point at every arrow [57]. A literature record may motivate a hypothesis; it must not become Θ [45,47,51,57].

The problem is therefore to keep those layers as typed objects, to refuse skip-level promotion (including LDHA as p_{lactate}), and to define scientific success as a staged, falsifiable chain rather than as disease eradication [11,12,36,43-52,56,58].

2. Justification of the Study

Existing integration habits fail in documented ways that this study is built to catch.

Overclaiming. GLOBOCAN 2024 estimates (about 20.6 million diagnoses and 9.8 million deaths) and WHO language that many cancers can be cured if found early and treated well are health-system context [1-4,54,55,64]. They are not CONFLUENCE parameters and not a product claim [56,59]. Adaptive-therapy and mathematical-oncology papers supply in-silico laboratories and control ideas; they are not CONFLUENCE efficacy [8,9,39-41]. Biomedical knowledge graphs integrate assertions; they remain literature objects until an identification step is performed [53].

Parameter smuggling. The P0 findings record the failure mode: treat OnCo's LDHA lactate-metabolism listing as the v2 symbol p_{lactate} , or treat validation/gene_to_parameter_map.json (LDHA $\rightarrow$ pyruvate_to_lactate (+0.10 / +0.30)) as identified Θ [10,11,36,58,63]. Alias trap: legacy pyruvate_to_lactate is not v2 p_{lactate} (default 0.22) [58,63]. Warburg and lactate-immune papers explain why the gene is scientifically interesting; they do not identify the coefficient [32-38,42]. OnCo confidence.probability is not $P(H)$. Idea maturity is not evidence level [10,57,59].

Missing gates. Structural identifiability asks whether a unique parameter vector is consistent with noise-free input-output data [45,46,48,50]. Practical identifiability asks whether finite, noisy data actually constrain those parameters [47,52]. Nonlinear biological systems are often only partially observable; unidentified symbols must remain labelled as such [51]. Without Gate 4, a silent write from a knowledge record into CancerODE produces a

confident but unidentifiable model [43-52,57]. Hallmarks, clonal evolution, intra-tumour heterogeneity, and the tumour microenvironment further explain why a named gene or a TNBC subtype label is not a sufficient statistic for Θ [6,7,13-21,22-31].

What this study therefore does. Pull request #9 shipped a read-only OnCo adapter and nine refusal tests without editing CancerODE [11,58]. This manuscript records those gates as architectural evidence, not as therapeutic efficacy [11,56].

3. Significance of the Study

Scientific significance for researchers. The work treats cancer as a dynamic, adaptive, microenvironment-coupled process rather than as a bag of independent named targets [7,13-17,22-27]. It gives laboratories a typed conversion ladder and a success criterion — traceability, mathematical validity, identifiability, out-of-sample prediction, experimental falsification — that can fail honestly [12,43-52,57]. Failed or refused claims (LDHA as p_{lactate}) stay visible [11,58,63].

Methodological significance. Evidence gates and the identifiability literature become refusal rules, not a new structural-identifiability proof for `confluence_v2_15d` [45-52]. Layer objects (OncoRef, Evidence, Mechanism, Parameter, Prediction) may point at the next layer; they may not collapse into it [57]. Public claims require a ledger id or a numbered Vancouver reference; DOIs are not invented [59].

What this significance is not. It is not clinical decision support, not a medical device, not a dosing table, and not a cure [3,56]. GLOBOCAN burden figures motivate computational oncology; they do not licence a CONFLUENCE product [1,2,56,64]. Translation waits for Gates 6–8, which this repository does not claim to have passed [3,12,56]. The significance of the study is methodological honesty for researchers, not a path to a clinic.

Specific aims

1. Forbid skip-level promotion from OnCo knowledge to ODE parameters [10,11,57,58].
2. Classify legacy artefacts, including `validation/gene_to_parameter_map.json`, as assumed / unidentified rather than as identified Θ [47,51,52,63].
3. Ship a read-only P0 adapter without editing CancerODE [11,58].
4. Keep the OnCo Ideas shelf separate from any CONFLUENCE hypothesis library [10,61].
5. Define success as gated, falsifiable computational work — control and disease-specific clearance — not a universal cure [3,12,56].

Background

Two artefacts, two questions

OnCo answers what is named, linked, dated and cited [10,61]. Biomedical knowledge graphs more generally integrate heterogeneous assertions; they remain literature objects until an identification step is performed [53]. CONFLUENCE v2 answers how a frozen 15-D state moves under an in-silico controller [11,58,60]. The working hypothesis of the findings chapter is that those artefacts are complementary provided they are not collapsed into each other [57].

Frozen lineages in this repository are:

ID — Role — Status

tnbc_mod_3s — TNBC-Metabolic-Strain-MOD notebooks — frozen; ROS audit pending [62]

confluence_report_6s — report architecture $X = [T, I, S, L, R, H]$ — paper only [60]

confluence_v2_15d — live 15-D CancerODE — adapter sits around it [11,58]

confluence_v1_calibrator — gene_to_parameter_map.json — executable, not identified [63]

OnCo integration does not justify changing the dynamics [11,58].

OnCo Ideas carry hypothesis, rationale, test and maturity fields [10,61]. Bulk ingest into CONFLUENCE is forbidden [57,58]. Idea maturity is not evidence level. OnCo confidence.probability is not $P(H)$ [57,59].

Hallmarks, heterogeneity and the microenvironment

Hanahan and Weinberg organised neoplastic growth as a small set of acquired capabilities, later expanded and re-dimensioned [13-15]. That organising frame is why a knowledge graph of named targets is not a closed dynamical explanation: the same hallmark can be implemented by different molecular routes [14,15,20]. Clonal evolution and ecological competition supply the time axis [16,17]. Intra-tumour heterogeneity is both a looking-glass for those processes and a documented source of therapeutic failure [7,18-21].

The tumour microenvironment is not a passive backdrop. Recruited stromal and immune populations regulate progression and metastasis [22-24]. T-cell exclusion and oncogenic-pathway-driven immune evasion are review-level mechanisms for why a metabolic or antigenic fact does not become a control law [25,26]. Exhaustion is a distinct T-cell state, not a synonym for “the immune variable is low” [27]. None of those reviews is treated here as an identified CONFLUENCE coupling.

Warburg metabolism and why LDHA is still not p_{lactate}

Aerobic glycolysis is a classical observation in cancer cells [32]. Later work reframed the Warburg effect as a proliferation-linked metabolic programme rather than a simple mitochondrial defect [33-35]. Lactate is not only a waste product: tumour-derived lactic acid can polarise macrophages and blunt T- and NK-cell surveillance [36-38]. A microenvironmental model of carcinogenesis treats acidification and selection as coupled processes [42].

OnCo lists LDHA among lactate-metabolism targets [10]. The CONFLUENCE v2 symbol p_{lactate} is a frozen-model parameter. The P0 adapter refuses to write the OnCo record into that symbol [11,36,58,63]. The literature explains why the gene is scientifically interesting. It does not identify Θ .

Mathematical oncology and adaptive control as research context

Ordinary differential equations, stochastic models, agent-based approaches and control-theoretic methods can represent tumour growth, treatment response and resistance as in-silico laboratories [8,39]. Adaptive therapy, introduced as an evolutionary alternative to continuous maximum-tolerated dosing, treats sensitive clones as competitive suppressors of resistant clones [9,40]. A later clinical-research programme in metastatic castrate-resistant prostate cancer is cited only as context that the control idea has a literature, not as a CONFLUENCE protocol or a patient result [41]. CONFLUENCE controllers that implement a Gatenby-style treat-and-halt heuristic remain in-silico research arms [9,12,56].

Layer objects (ontology spec v0.3)

The ingestion spec keeps five objects [57]. A record may point at the next layer. It may not collapse into it.

Layer — Object — Allowed question — Forbidden leap

Knowledge — OncoRef — What does OnCo name and link? — therefore $k = \dots$

Evidence — EvidenceObject — What was measured, where, in what system? — therefore this term belongs in F

Causal mechanism — MechanismObject — do(U) changes which state, in which context? — therefore identifiable from CCLE

Parameter — ParameterObject — Which symbol in which frozen model? — silent write into rhs_cancer

Prediction — PredictionObject — Frozen-model output under U, with uncertainty — clinical advice

Conversion ladder

```
` OnCo knowledge --cite--> Evidence (source URI + rung required) --interpret--> Hypothesis (falsifier required)
--propose--> Mechanism (context + sign + do-operator) --identify--> Parameter (model_id + symbol + identifiability
!= unidentified) --simulate--> Prediction --test--> Experiment --write--> Evidence `
```

Every arrow is a failure point. A page may motivate a hypothesis; it must not become Θ [45,47,51,57]. No arrow may skip a box. Wired RHS terms remain Parameters with provenance assumed until identified [52,63].

legacy gene $\rightarrow$ parameter map $\neq$ identified parameter mapping. Alias trap: legacy pyruvate_to_lactate is not v2 p_lactate (default 0.22) [58,63].

Methods

Honesty gates

The public protocol is sequential and falsifiable [12,56,59]. It is not a treatment path.

Gate — Requirement

0 — Research-only scope. Not a medical device. Not clinical decision support. [56]

1 — Provenance: source, URI, date, population. [59]

2 — Evidence class named. [57]

3 — Mechanism: intervention, state, sign, context, falsifier. [57]

4 — Parameter: model id, symbol, units, estimator, uncertainty, identifiability. Unidentified $\neq$ established. [45-52]

5 — Prediction labelled as computational. [56]

6 — Held-out / external / baseline validation. [12]

7 — An experiment that could show the prediction is wrong. [12]

8 — Translation only after biological and clinical validation this repository does not claim. [3,56]

Scientific success is: traceability → mathematical validity → identifiability → out-of-sample prediction → experimental falsification [43,44,51,52].

Identifiability as a methods claim, not a result

Systems biology treats a model as a formal object whose parameters may or may not be recoverable from observations [43,44]. Structural identifiability asks whether a unique parameter vector is consistent with noise-free input–output data [45,46,48,50]. Practical identifiability asks whether finite, noisy data actually constrain those parameters [47,52]. Nonlinear biological systems are often only partially observable; unidentified symbols must remain labelled as such [51]. CONFLUENCE Gate 4 is that literature applied as a refusal rule: OnCo confidence and the legacy numeric map are not estimators [10,47,63]. This manuscript does not report a new identifiability computation.

Adapter P0

confluence/onco/ is a read-only client: cache envelope, bindings, schemas [11,58]. Commands documented in docs/ONCO_ADAPTER.md include fixture meta, bind --id ldha, and wired [58]. bind() returns slot annotations. It never writes p_lactate or pyruvate_to_lactate [11,63]. refuse_knowledge_as_parameter returns provenance forbidden [11,57]. Tests live in tests/test_onco_adapter.py [11,60].

OnCo data is CC BY-NC 4.0 [10,61]. Adapter code in this repository is MIT [60]. Cached payloads are not vendored; live cache stays under data/onco/cache/ (gitignored) [58]. Attribution on every export: Data from OnCo (onco.cc), CC BY-NC 4.0; commercial use needs a licence [10,59].

P0 ships: read-only client + cache envelope + bindings + nine gates [11,58]. P0 does not ship: RHS edits, controller edits, 3-state ROS import, 6-state expansion, automatic fitting, reverse writes to OnCo [11,58,62].

Refusal rules

1. Do not vendor the OnCo corpus [10,61].
2. OnCo is not a parameter source. refuse_knowledge_as_parameter(OncoRef("ldha"), "p_lactate") returns provenance forbidden [11,36,58].
3. OnCo is not a controller prior [9,40,58].
4. Wired RHS terms remain Parameters with provenance assumed until identified [47,52,63].
5. No reverse writes to OnCo [10,61].
6. No ODE right-hand-side edits in the P0 / thesis-#1 scope [11,58].
7. Public claims need a ledger id or a numbered reference (docs/CITATION_POLICY.md) [59].
8. Do not fabricate DOIs [59].

This manuscript uses the Vancouver entries generated from docs/manuscript/thesis_01_bibliography.json. Every doi: field was checked against Crossref or PubMed on 20 September 2026. Agency and GitHub items are URL citations without a DOI [2-7,10-12,56-64].

What was not done

No patient data. No ODE refit. No automatic fitting. No 3-state ROS import. No 6-state expansion. No clinical trial, RECIST adjudication, or dosing table [12,56].

Results / architectural findings

These are specification and software-architecture findings. They are not experimental oncology results and not patient outcomes [56].

1. **Honesty ladder holds as an object model.** Knowledge, Evidence, Causal mechanism, Parameter and Prediction are separate types [57]. A record may point at the next layer. It may not collapse into it.
2. **LDHA is not `p_lactate` (refuse-as-parameter).** `refuse_knowledge_as_parameter(OncoRef("ldha"), "p_lactate")` is forbidden [11,58]. The auditor demo claim — that LDHA expression can be entered as `p_lactate` because OnCo lists LDHA as a lactate-metabolism target — is the claim the gate is built to reject [10,32,36,63].
3. **Legacy map is assumed / unidentified.** `validation/gene_to_parameter_map.json` maps LDHA → `pyruvate_to_lactate (+0.10 / +0.30)` [63]. That row is a legacy numeric hint, not an identified v2 parameter [47,51,52].
4. **No ODE writes.** PR #9 and the P0 adapter do not edit `CancerODE.rhs_cancer` [11,58].
5. **OnCo is a cited graph, not Θ [10,61].** Counts move with `buildDate`.
6. **Three-arm programme remains a protocol.** `validation/validation_protocol.md` states H0 versus H1. Thresholds such as $r > 0.5$ are criteria, not findings [12].
7. **Public ledger keeps year and population.** Claims EVID-001–EVID-010 on the thesis page cite [1-12] without promoting burden statistics into model parameters [1,5,59,64].

No wet-lab measurement is reported here. No new parameter was identified [45,52,56].

Discussion

Keeping OnCo and CONFLUENCE apart is the result [10,11,57]. Integration is valuable only as a gated pipeline: cite → interpret → propose → identify → simulate → test [49,57]. Collapsing any two boxes produces a confident but unidentifiable model [45,47,51].

The hallmarks, clonal-evolution and microenvironment literatures explain why a named gene, a lactate observation, or a TNBC subtype label is not a sufficient statistic for Θ [13-15,16-21,22-27,28-31]. Warburg and lactate-immune papers explain why LDHA appears in OnCo and why lactate is biologically non-trivial; they still do not license a silent write into `p_lactate` [32-38,42]. Mathematical oncology and adaptive-therapy papers explain why a controller layer can exist around a frozen model; they are not CONFLUENCE efficacy [8,9,39-41].

WHO language that many cancers can be cured if found early and treated well is retained as a health-system statement [3,4]. It is not transferred to CONFLUENCE as a product claim [56]. Prior GLOBOCAN editions (2020, 2022) are cited so that the 2024/2026 series is not treated as a one-off number [1,54,55,64].

Disease profiles and the thinking lab are scaffolds for questions. They are not protocols and are not the Scholar landing page [56,59]. The Scholar article URL is `/thesis` with the same-directory PDF `/thesis.pdf`.

Do not expand the state vector merely because OnCo lists more cell types [10,23,61]. Cache has no TTL yet [58]. Those are engineering limits, not clinical limits [56].

Limitations

- No new wet-lab measurement [56]. - PR #9 does not identify p_{lactate} or any other Θ [11,52]. - The 3-state ROS notebook remains unaudited [62]. - The three-arm protocol can fail; that would still be a result [12]. - OnCo counts move with buildDate [10]. - The optional Grok auditor may miss sources; “insufficient” is preferred to invented citations [59]. - Nigeria 2022 estimates are setting context only [5]. - This document has no DOI. - Identifiability citations define Gate 4; they are not a new structural-identifiability proof for confluence_v2_15d [45-52].

Future work

1. Iterate the adapter without RHS edits [11,58].
 2. Admit-list a hypothesis library separate from OnCo Ideas [10,61].
 3. Identify p_{lactate} only after gates 1–4 stay green and an estimator with uncertainty exists [47,52,58].
 4. External / held-out validation (Gate 6) before any experimental falsifier (Gate 7) [12].
 5. Optional later: deposit this PDF on Zenodo or a preprint server and only then add citation_doi to the HTML landing page.
- Translation (Gate 8) is out of scope until biological and clinical validation that this repository does not claim [3,56].

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